

FR3K: Master Strategic Implementation Plan

Free Communications. Open Infrastructure. AI Driven. Community Built.

Version: 1.1

Date: August 2026

Status: Implementation Baseline

1. Executive Summary

FR3K is an engineering-led initiative focused on building resilient, community-owned communications infrastructure in North Queensland. The project prioritizes **transparent experimentation, reproducible evidence, and regulatory compliance**.

The core strategic thesis is that by documenting both successes and failures in a difficult tropical environment, FR3K builds the trust necessary to attract contributors, partners, and sustainable funding.

2. Strategic Risk & Mitigation

To ensure the long-term viability and safety of the project, the following primary risks have been identified along with their corresponding mitigation strategies.

Risk Category	Primary Threat	Mitigation Strategy	Accountable
Regulatory	ACMA non-compliance or interference complaint.	Mandatory RF Lead sign-off for every unique radio/antenna configuration; documented EIRP/PSD calculations.	RF Lead
Legal	Unauthorised site access or liability from equipment failure.	Written site permissions (SOP-06); clear “supplementary infrastructure” disclaimers; robust insurance coverage.	Operations Lead
Technical	Hardware failure due to tropical corrosion or thermal stress.	Marine-grade enclosures; thermal testing under peak load; conservative solar yield modeling (SOP-08).	Engineering Lead
Safety	Team injury during field deployment or battery incident.	Mandatory safety briefings; “Stop Work” authority for all members; LiFePO4 chemistry for unattended nodes.	Safety Lead
Community	Misinformation or loss of trust due to unsupported claims.	“Evidence-first” publishing; mandatory human review of all AI content; transparent failure reporting.	Program Lead

3. Technical Architecture & Regulatory Guardrails

3.1 Layered Network Design

FR3K employs a layered architecture to ensure flexibility and resilience:

- **Meshtastic:** Accessible, handheld LoRa ecosystem for field teams and introductory workshops.
- **Reticulum/LXMF:** The principal research architecture for identity, resilient messaging, and multi-interface routing.

- **Backhaul:** Agnostic support for Ethernet, Wi-Fi, and satellite (e.g., Starlink) where appropriate.

3.2 ACMA Compliance Checklist

All FR3K transmissions must adhere to the **Radiocommunications (Low Interference Potential Devices) Class Licence 2025**.

Requirement	Verification Step	Status
Frequency	Confirmed within 915–928 MHz band.	[]
Modulation	Digital modulation or frequency hopping verified.	[]
Power	Conducted power and antenna gain result in compliant EIRP.	[]
PSD	Peak Power Spectral Density within 8 dBm/3 kHz (if applicable).	[]
Hardware	Equipment carries required compliance markings.	[]
Sign-off	RF Lead has reviewed and signed the compliance record.	[]

4. 90-Day Launch Roadmap

The first 90 days focus on establishing a credible technical and operational baseline.

- 1. Phase 1: Bench & Baseline (Days 1-30)**
 - Finalize P0 Bench Reference design.
 - Complete initial ACMA compliance audit for all core hardware.
 - Establish documentation and media pipelines.
- 2. Phase 2: Controlled Pilots (Days 31-60)**
 - Deploy first P1 Portable Field Kits for range surveys.
 - Secure first two private host sites for P2 nodes.
 - Launch the “Scout” volunteer onboarding pathway.
- 3. Phase 3: Community Exercise (Days 61-90)**
 - Conduct first governed community messaging exercise.
 - Publish “Month 2” KPI and evidence report.
 - Review and refine architecture for regional expansion.

5. AI & Hermes Governance

AI is used as an **amplifier**, not a replacement for human judgment.

- **Traceability:** Every AI-generated asset must link to its prompt, model, and human reviewer.
 - **Honesty:** Conceptual imagery must be clearly labeled to avoid confusion with real deployment evidence.
 - **Safety:** AI is never permitted in safety-critical control loops.
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6. Conclusion

FR3K is not just a network of nodes; it is a network of people. By maintaining a disciplined, evidence-based approach, we provide a blueprint for regional resilience that can be reproduced by communities across Australia and beyond.